

Propositional Equivalences

Section 1.3
partB

Constructing New Logical Equivalences

- We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.
- To prove that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B.

$$\begin{array}{c} A \equiv A_1 \\ \vdots \\ A_n \equiv B \end{array}$$

- Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$$\begin{aligned}
 \neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the second De Morgan law} \\
 &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the first De Morgan law} \\
 &\equiv \neg p \wedge (p \vee \neg q) && \text{by the double negation law} \\
 &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the second distributive law} \\
 &\equiv F \vee (\neg p \wedge \neg q) && \text{because } \neg p \wedge p \equiv F \\
 &\equiv (\neg p \wedge \neg q) \vee F && \text{by the commutative law for disjunction} \\
 &\equiv (\neg p \wedge \neg q) && \text{by the identity law for } \mathbf{F}
 \end{aligned}$$

Equivalence Proofs

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology.

Solution:

$$\begin{aligned}
 (p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by truth table for } \rightarrow \\
 &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the first De Morgan law} \\
 &\equiv (\neg p \vee p) \vee (q \vee \neg q) && \text{by associative and commutative laws} \\
 &\equiv T \vee T && \text{laws for disjunction} \\
 &\equiv T && \text{by truth tables} \\
 &&& \text{by the domination law}
 \end{aligned}$$

Logical Equivalences

Examples:

1. Show that $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$ are not logically equivalent.

Hint:

To show these are not logically equivalent, we need only find **one assignment** of truth values to p, q, r, s for which the truth values of the two propositions differ.

Dual (对偶式)

- The dual of compound proposition that contains only the logical operators $\vee, \wedge$ and $\neg$ is the proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F and each F by T. the dual of S is denoted by S^* .
- $S = (p \vee \neg q) \wedge r \vee T$ $S^* = (p \wedge \neg q) \vee r \wedge F$

Dual (cont...)

- $S = (p \wedge q) \rightarrow (p \vee q) = \neg(p \wedge q) \vee (p \vee q)$
- $S^* = \neg(p \vee q) \wedge (p \wedge q)$
- Theorem: let s and t are two compound propositions, $s \Leftrightarrow t$ if and only if $s^* \Leftrightarrow t^*$

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Exercises

- Find a compound proposition involving the propositional variables p, q, r that is true when exactly two of p, q , and r are true and is false otherwise.

Solution:

$$p \wedge q \wedge \neg r$$

$$p \wedge \neg q \wedge r$$

$$\neg p \wedge q \wedge r$$

$$(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$$

Exercises

- The proposition $p \text{ NOR } q$ is true when both p and q are false, and it is false otherwise. The operator $\downarrow$ is called **Peirce arrow**. (或非)
 - (a) show that $p \downarrow p$ is logically equivalent to $\neg p$
 - (b) show that $(p \downarrow q) \downarrow (p \downarrow q)$ is logically equivalent to $p \vee q$
- The proposition $p \text{ NAND } q$ is true when either p or q , or both, are false; and it is false when both p and q are true. The operator $|$ is called **Sheffer stroke**. (与非)

Functionally complete operators

- A collection of logical operators is called functionally complete if every compound proposition is logically equivalent to a compound proposition involving only these logical operators.
- $\{\neg, \vee\}, \{\neg, \wedge\}, \{|\}, \{\downarrow\}$ are all functionally complete operators.

Propositional Satisfiability

- A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.
- A compound proposition is unsatisfiable if and only if it is a **contradiction** or its **negation** is a **tautology**.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

Solution: Satisfiable. Assign **T** to p , q , and r .

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Satisfiable. Assign **T** to p and **F** to q .

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Not satisfiable. Check each possible assignment of truth values to the propositional variables and none will make the proposition true.

Notation

$\bigvee_{j=1}^n p_j$ is used for $p_1 \vee p_2 \vee \dots \vee p_n$

$\bigwedge_{j=1}^n p_j$ is used for $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Needed for the next example.

Sudoku

- A **Sudoku puzzle** is represented by a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**. Some of the 81 cells of the puzzle are assigned one of the numbers 1, 2, ..., 9.
- The puzzle is solved by assigning numbers to each blank cell so that every row, column and block contains each of the nine possible numbers.
- Example

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem

- Let $p(i,j,n)$ denote the proposition that is true when the number n is in the cell in the i th row and the j th column.
- There are $9 \times 9 \times 9 = 729$ such propositions.
- In the sample puzzle $p(5,1,6)$ is true, but $p(5,j,6)$ is false for $j = 2,3,\dots,9$

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1				9			
							6	

Encoding (cont)

- For each cell with a given value, assert $p(i,j,n)$, when the cell in row i and column j has the given value.
- Assert that every row contains every number.

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i,j,n)$$

- Assert that every column contains every number.

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i,j,n)$$

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1				9			
							6	

Encoding (cont)

- Assert that each of the 3×3 blocks contain every number.

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigvee_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$$

(this is tricky - ideas from chapter 4 help)

- Assert that no cell contains more than one number. Take the conjunction over all values of n, n', i , and j , where each variable ranges from 1 to 9 and $n \neq n'$, of

$$p(i, j, n) \rightarrow \neg p(i, j, n')$$

Solving Satisfiability Problems

- To solve a Sudoku puzzle, we need to find an assignment of truth values to the 729 variables of the form $p(i, j, n)$ that makes the conjunction of the assertions true. Those variables that are assigned T yield a solution to the puzzle.
- A truth table can always be used to determine the satisfiability of a compound proposition. But this is too complex even for modern computers for large problems.
- There has been much work on developing efficient methods for solving satisfiability problems as many practical problems can be translated into satisfiability problems.

Homework

- 第8版 Sec. 1.3 34, 36, 44, 55

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Section 1.3 (supplement)

Propositional Normal Forms (命题范式)

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Propositional Formula

- (1) Each propositional variable is a formula, and the truth values T and F are also formulas.
- (2) If A is formula, so is $(\neg A)$.
- (3) If A and B are formulas, so are $(A \vee B)$, $(A \wedge B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$.
- (4) A string of symbols is a formula only as it is determined by (finitely many applications of) the above three rules.

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Example

- **EXAMPLE**
- The following string are formulas:
 - $\neg(p \vee q)$, $p \rightarrow (q \rightarrow r)$, $(p \wedge q) \rightarrow r$
- The following string are not formulas:
 - $pq \rightarrow r$, $\neg p \rightarrow q) \rightarrow r$

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Propositional Normal Forms

- Formulas can be transformed into **standard forms** so that they become more convenient for symbolic manipulations and make identification and comparison of two formulas easier.

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Normal Forms (范式)

- There are two types of normal forms in propositional calculus:
- the **disjunctive normal form(DNF)** (析取范式)
- and
- the **conjunctive normal form(CNF)** (合取范式) .

DEFINITION. A **literal**(文字) is a variable or its negation

Example p and $\neg p$ are literals, but $p \wedge q$ is not.

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Propositional Normal Forms

- DEFINITION. A formula is said to be in **disjunctive normal form** if it is written as a disjunction, in which all the terms are conjunctions of literals.
- EXAMPLE 1 $(p \wedge q) \vee (p \wedge \neg q)$, $p \vee (q \wedge r)$, $\neg p \vee T$ are in disjunctive normal forms. The disjunction $\neg(p \wedge q) \vee r$ is not in normal form. (Why?)

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Clauses (子句)

- In general a formula in disjunctive normal form is
- $(A_{11} \wedge \dots \wedge A_{1n}) \vee \dots \vee (A_{k1} \wedge \dots \wedge A_{kn})$
- DEFINITION. Disjunctions (conjunctions) with literals as disjuncts (conjuncts) are called disjunctive (conjunctive) clauses. Disjunctive and conjunctive clauses are simply called clauses.

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Clauses (cont...)

Example: $p \wedge q \wedge \neg r$ is a conjunctive clause

$\neg p \vee q \vee \neg r$ is a disjunctive clause

Conjunctive clause is also called **basic product** and

Disjunctive clause is also called **basic addition**.

DEFINITION. A conjunction with disjunctive clauses as its conjunctions is said to be in **Conjunctive normal form**.

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Cont..

EXAMPLE 2 $p \wedge (q \vee r)$ and $p \wedge F$ are in conjunctive normal form.

However $p \wedge (r \vee (p \wedge q))$ is not in conjunctive form.

In general a formula in conjunctive normal form is

$$(A_{11} \vee \dots \vee A_{1n}) \wedge \dots \wedge (A_{k1} \vee \dots \vee A_{knk})$$

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Conjunctive normal form

Observe the following formulas:

- (1) p
- (2) $\neg p \vee q$
- (3) $\neg p \wedge q \wedge \neg r$
- (4) $\neg p \vee (q \wedge \neg r)$
- (5) $\neg p \wedge (q \vee \neg r) \wedge (\neg q \vee r)$

(1),(2),(3),(5) are in conjunctive normal forms.

(1),(2),(3),(4) are in disjunctive normal forms

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How to obtain normal forms

(1) Eliminating $\rightarrow, \leftrightarrow, \mid, \downarrow$

(2) Eliminating $\neg, \vee, \wedge$ from the scope of $\neg$, such that any $\neg$ has only an atom as its scope. In other words eliminating the parenthesis following behind $\neg$

This method leads to obtaining the disjunctive or conjunctive normal forms.

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Example

Convert the following formula into a conjunctive normal form.

$$\neg((p \vee \neg q) \wedge \neg r)$$

The conjunctive normal form can be found by the following derivations:

$$\neg((p \vee \neg q) \wedge \neg r)$$

$$(1) \neg(p \vee \neg q) \vee (\neg \neg r) \quad \text{De Morgan}$$

$$(2) \neg(p \vee \neg q) \vee r \quad \text{Double negation}$$

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Example

$$(3) (\neg p \wedge \neg \neg q) \vee r \quad \text{De Morgan}$$

$$(4) (\neg p \wedge q) \vee r \quad \text{Double negation}$$

$$(5) (\neg p \vee r) \wedge (q \vee r) \quad \text{Distributivity}$$

EXAMPLE 5 Convert the following formula into a conjunctive normal form.

$$(p_1 \wedge p_2) \vee (p_3) \wedge (p_4 \vee p_5)$$

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Example

- Solution:
- $(p_1 \wedge p_2) \vee (p_3) \wedge (p_4 \vee p_5) \Leftrightarrow$
- $(p_1 \wedge p_2) \vee (p_3 \wedge (p_4 \vee p_5)) \Leftrightarrow$
- $((p_1 \wedge p_2) \vee p_3) \wedge ((p_1 \wedge p_2) \vee (p_4 \vee p_5))$
- $\Leftrightarrow (p_1 \vee p_3) \wedge (p_1 \vee p_4 \vee p_5) \wedge (p_2 \vee p_3) \wedge (p_2 \vee p_4 \vee p_5)$

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• Examples

2. Convert the following formula into conjunctive and disjunctive normal forms. $\neg(p \vee q) \leftrightarrow (p \wedge q)$

Solution:

$$\begin{aligned}
 & \neg(p \vee q) \leftrightarrow (p \wedge q) \\
 & \equiv (\neg(p \vee q) \rightarrow (p \wedge q)) \wedge ((p \wedge q) \rightarrow \neg(p \vee q)) \\
 & \equiv ((p \vee q) \vee (p \wedge q)) \wedge (\neg(p \wedge q) \vee \neg(p \vee q)) \\
 & \equiv ((p \vee q \vee p) \wedge (p \vee q \vee q)) \wedge ((\neg p \vee \neg q) \vee (\neg p \wedge \neg q)) \\
 & \equiv (p \vee q) \wedge (\neg p \vee \neg q \vee \neg p) \wedge (\neg p \vee \neg q \vee \neg q) \\
 & \equiv (p \vee q) \wedge (\neg p \vee \neg q)^* \\
 & \equiv ((p \vee q) \wedge \neg p) \vee ((p \vee q) \wedge \neg q) \\
 & \equiv (p \wedge \neg p) \vee (q \wedge \neg p) \vee (p \wedge \neg q) \vee (q \wedge \neg q) \\
 & \equiv (q \wedge \neg p) \vee (p \wedge \neg q)^{**}
 \end{aligned}$$

Remark: Formulas (*) is in CNF and formulas (**) is in DNF.

Disjunctive normal forms from truth tables

- So far we have shown how to find the truth table of a logical formula. The reverse is also possible. One can convert any given truth table into a formula.
- The formula obtained in this way is already in disjunctive normal form.
- In fact, the conceptually easiest method to find the normal form of a formula is by using truth tables. Unfortunately, truth tables grow exponentially with the number of variables,

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Disjunctive normal forms from truth tables

- which makes this method unattractive for formulas with many variables.
- Generally, a truth table gives truth values of some formula for all possible assignments. The table below gives an example of truth table for a certain formula f . The truth values of f depends on the three variables p, q, r .
- This makes f a truth function, or a Boolean function with arguments

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(cont...)

- p q r f
- T T T T
- T T F F
- T F T T
- T F F F
- F T T F
- F T F F
- F F T T
- F F F F

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minterm

- To convert a Boolean function into a formula, one makes use of minterms.
- DEFINITION. A minterm is a conjunction of literals in which each variable is represented exactly once.
- EXAMPLE 7 If a Boolean function has the variables p, q, r then $p \wedge \neg q \wedge r$ is a minterm but $p \wedge \neg q$ and $p \wedge p \wedge \neg q$ are not.

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Representation of minterm

- There are 2^n different minterm for n propositional variables. For example there 4 different minterm for p, q , they are $p \wedge q, p \wedge \neg q, \neg p \wedge q, \neg p \wedge \neg q$. For the sake of simplification, we use m_j denote the minterms. Where j is a integer, its binary representation corresponds the evaluation of variables that make m_j be equal to T.

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Example

- For three propositional variables,
- $m_0 = \neg p \wedge \neg q \wedge \neg r$ $m_1 = \neg p \wedge \neg q \wedge r$
- $m_2 = \neg p \wedge q \wedge \neg r$ $m_3 = \neg p \wedge q \wedge r$
- $m_4 = p \wedge \neg q \wedge \neg r$ $m_5 = p \wedge \neg q \wedge r$
- $m_6 = p \wedge q \wedge \neg r$ $m_7 = p \wedge q \wedge r$
- If a propositional form is denoted by: $f = m_j \vee m_k \vee \dots \vee m_l$ we simply denote $f = \sum m(j, k, \dots, l)$

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Properties of minterms

- (1) Each minterm is true for exactly one assignment.
For example, $p \wedge \neg q \wedge r$ is true if p is T, q is F and r is T.
Any deviation from this assignment would make this particular minterm false.
- (2) The conjunction of two different minterm is always false. I.e. $m_i \wedge m_k = F$
- (3) The disjunction of all minterm is T
- $m_0 \vee m_1 \vee \dots \vee m_{2^n-1} = T$

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Properties of minterms

- A disjunction of minterms is true only if at least one of its constituents minterms is true. For example,
- $(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$
- is only true if at least one of $p \wedge q \wedge r$, $p \wedge \neg q \wedge r$ or $\neg p \wedge \neg q \wedge r$ is true.

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Minterms and truth tables

- If a function, such as f , is given by truth table, we know exactly for which assignments it is true. Consequently, we can select the minterms that make the function true and form the disjunction of these minterms.
- The function f , for instance, is true for three assignments:

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Disjunctive normal forms from truth tables

- 1. p, q, r are all true.
- 2. $p, \neg q, r$ are all true.
- 3. $\neg p, \neg q, r$ are all true.
- The disjunction of the corresponding minterms is tautologically equivalent to f , which means that we have the following formula for f :
- $f \Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$

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Full disjunctive form

- Actually, we have a special type of normal form, the full disjunctive form.
- DEFINITION. If a Boolean function is expressed as a disjunction of minterms, it is said to be in full disjunctive form.

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How to find Full disjunctive form

- Two methods:
- (1) From truth table;
- (2) From disjunctive form, changing its clauses to minterms.
- Example finding the From disjunctive form of formula
- $f = (p \wedge \neg q) \rightarrow (\neg r \wedge \neg p)$

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Solution

p	q	r	$p \wedge \neg q$	$\neg r \wedge \neg p$	$(p \wedge \neg q) \rightarrow (\neg r \wedge \neg p)$
F	F	F	F	T	T
F	F	T	F	F	T
F	T	F	F	T	T
F	T	T	F	F	T
T	F	F	T	F	F
T	F	T	T	F	F
T	T	F	F	F	T
T	T	T	F	F	T

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Cont..

- So $f = \sum m(0,1,2,3,6,7)$
- Solution 2:
- $f = (p \wedge \neg q) \rightarrow (\neg r \wedge \neg p)$
- $= \neg(p \wedge \neg q) \vee (\neg r \wedge \neg p)$
- $= \neg p \vee q \vee (\neg r \wedge \neg p)$
- $= (\neg p \wedge T \wedge T) \vee (q \wedge T \wedge T) \vee (\neg r \wedge \neg p \wedge T)$

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Cont..

- $= (\neg p \wedge (q \vee \neg q) \wedge (r \vee \neg r))$
- $\vee (q \wedge (p \vee \neg p) \wedge (r \vee \neg r))$
- $\vee (\neg r \wedge \neg p) \wedge (q \vee \neg q)$
- $= (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee$
- $(\neg p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$
- $= m_0 \vee m_1 \vee m_2 \vee m_3 \vee m_6 \vee m_7$

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Conjunctive Normal Form

- A compound proposition is in *Conjunctive Normal Form* (CNF) if it is a conjunction of disjunctions.
- Every proposition can be put in an equivalent CNF.
- Conjunctive Normal Form (CNF) can be obtained by eliminating implications, moving negation inwards and using the distributive and associative laws.
- Important in resolution theorem proving used in artificial Intelligence (AI).
- A compound proposition can be put in conjunctive normal form through repeated application of the logical equivalences covered earlier.

Conjunctive Normal Form

Example: Put the following into CNF:

$$\neg(p \rightarrow q) \vee (r \rightarrow p)$$

Solution:

1. Eliminate implication signs:

$$\neg(\neg p \vee q) \vee (\neg r \vee p)$$

2. Move negation inwards; eliminate double negation:

$$(p \wedge \neg q) \vee (\neg r \vee p)$$

3. Convert to CNF using associative/distributive laws

$$(p \vee \neg r \vee p) \wedge (\neg q \vee \neg r \vee p)$$

Conjunctive Normal Form and Disjunctive form

- We can convert Full Disjunctive Normal form to Full Conjunctive Normal Form directly

$$\text{Let } f = \sum m(j, k, \dots, l), \quad g = \sum m(\{0, 1, 2, \dots, 2^{n-1}\} - \{j, k, \dots, l\})$$

Then $f \vee g = T$, $f \wedge g = F$, hence

$$f = \neg g = \pi M(\{0, 1, 2, \dots, 2^{n-1}\} - \{j, k, \dots, l\})$$

Where M_i is $\neg m_i$ is a disjunction.

(Maxterm 极大项)

Homework

- Find the Full Disjunctive (Conjunctive) Normal Form from truth table of following formula
 $((\neg p \wedge q) \rightarrow (p \leftrightarrow r)) \vee (q \rightarrow \neg r)$